

FBISE BIOLOGY - CLASS 9

Comprehensive Guess Paper for 2nd Annual Examination 2026

Examination date 25 September 2026	Current theory pattern 12 MCQs + 10 Short + 3 Long	Total theory marks 60 marks
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What this pack contains	36 high-yield MCQs, 30 short-question choices, 6 core long-question choices, 6 reserve long questions, an answer key, and a diagram checklist.
Syllabus control	Every included question is drawn only from SLOs marked Summative in the uploaded FBISE assessment framework.
Paper evidence	Calibrated against the 2025 model paper, 2025 annual papers, 2025 second annual paper, and all three 2026 first annual papers.
Important exclusion	The detailed observations made by Darwin during the HMS Beagle voyage (SLO B-09-B-04) are formative-only and are not included as a direct annual-exam question.

How students should use it: First master the Core Guess (Questions 1-12, the first 10 short-question pairs, and the three long-question pairs). Then prepare the reserve questions. For every diagram, practise title, correct labels, neat pencil lines and the related written explanation.

Academic note: This is an evidence-based preparation resource, not an official paper or a guarantee of exact questions. Students should still revise all summative SLOs.

SECTION A - MOST EXPECTED MCQs

Questions 1-12 form the Core Guess. Questions 13-36 are high-value reserve MCQs.

No.	Question and options
1	A statement proposed on the basis of observations and testable through experiments is called: A. law B. hypothesis C. theory D. deduction
2	Which outcome best demonstrates scientific collaboration? A. one biologist works alone B. data are kept secret C. different specialists share and critique evidence D. results are accepted without review
3	The scientific study of organisms and their evolutionary relationships is: A. classification B. taxonomy C. systematics D. binomial nomenclature
4	A group of organisms that naturally interbreed and produce fertile offspring is a: A. genus B. species C. population D. community
5	Which evidence of evolution includes homologous and vestigial structures? A. palaeontology B. comparative anatomy C. selective breeding D. biogeography
6	A family is a group of related: A. species B. genera C. orders D. classes
7	Which kingdom contains multicellular, heterotrophic eukaryotes with chitinous cell walls? A. Animalia B. Plantae C. Fungi D. Protista
8	Why are viruses difficult to classify as living organisms? A. they contain no protein B. they show both living and non-living features C. they are all harmful D. they lack genetic material
9	The monomers of proteins are: A. fatty acids B. monosaccharides C. amino acids D. nucleotides
10	Which combination forms a typical lipid molecule? A. amino acids and glycerol B. glucose and glycerol C. fatty acids and glycerol D. glucose and fatty acids
11	The nitrogenous bases between the two DNA strands are joined by: A. ionic bonds B. hydrogen bonds C. peptide bonds D. glycosidic bonds
12	If a DNA template has the sequence A-T-C-G-T, the complementary mRNA sequence is: A. U-A-G-C-A B. T-A-G-C-A C. U-T-G-C-U D. A-U-C-G-U
13	Which organelle is absent from a typical plant cell? A. centriole B. Golgi apparatus C. mitochondrion D. ribosome
14	Which organelle is known as the protein factory of the cell? A. lysosome B. nucleus C. ribosome D. mitochondrion
15	DNA replication occurs during: A. G1 phase B. S phase C. G2 phase D. M phase
16	The longest major stage of the cell cycle is: A. prophase B. interphase C. metaphase D. cytokinesis
17	Meiosis maintains chromosome number across generations because it produces: A. diploid body cells B. haploid gametes C. identical cells D. polyploid cells
18	The stomach tissue mainly responsible for secreting gastric juice is: A. connective B. nervous C. muscular D. epithelial

No.	Question and options
19	An enzyme is best described as a: A. lipid molecule B. biological catalyst C. source of energy D. product of respiration
20	A permanently attached non-protein component of an enzyme is a: A. coenzyme B. activator C. prosthetic group D. substrate
21	A competitive inhibitor usually binds to the enzyme at its: A. allosteric site B. active site C. product site D. cofactor
22	ATP is called the energy currency because it: A. stores hereditary information B. provides readily usable energy C. makes cell walls D. carries oxygen
23	The final electron acceptor in aerobic respiration is: A. carbon dioxide B. oxygen C. NADH D. glucose
24	When oxygen is scarce, human muscle cells obtain energy mainly by: A. alcoholic fermentation B. lactic acid fermentation C. photosynthesis D. Krebs cycle
25	Light-dependent reactions of photosynthesis occur mainly in the: A. stroma B. thylakoid membranes C. cytoplasm D. nucleus
26	The principal food-conducting tissue in plants is: A. xylem B. phloem C. epidermis D. mesophyll
27	Strong wind generally increases transpiration by: A. closing stomata B. removing the humid air around leaves C. reducing leaf temperature only D. stopping diffusion
28	Magnesium is essential for the formation of: A. cellulose B. chlorophyll C. protein D. DNA
29	Which is a micronutrient required by plants? A. nitrogen B. phosphorus C. magnesium D. zinc
30	A plant cell normally does not burst in a hypotonic solution because of its: A. nucleus B. cell wall C. large nucleus D. Golgi apparatus
31	Producing genetically identical plants from an explant is called: A. grafting B. tissue culture C. pollination D. parthenogenesis
32	Which is an artificial method of vegetative propagation? A. binary fission B. spore formation C. grafting D. budding in yeast
33	In flowering plants, the zygote develops into the: A. embryo B. endosperm C. seed coat D. ovule
34	Double fertilization produces a zygote and: A. pollen tube B. endosperm nucleus C. seed coat D. stigma
35	Which structure is NOT a part of the embryo in a seed? A. radicle B. plumule C. cotyledon D. endosperm
36	The female gametophyte of a flowering plant is the: A. pollen grain B. embryo sac C. ovary D. anther

SECTION B - SHORT-QUESTION MASTER GUESS

Each item is worth 3 marks. The first ten pairs mirror the current paper; the last five pairs are reserves.

No.	Question	OR	Alternative question
1	Explain the steps of the scientific method in correct sequence. Distinguish hypothesis, theory and law.	OR	Explain how biology connects with any three other natural sciences, giving one example of each.
2	State Darwin's theory of evolution by natural selection and list its main points.	OR	Explain any three sources of variation that may lead to speciation and evolution.
3	Explain how palaeontology and comparative anatomy provide evidence of evolution.	OR	Write the taxonomic hierarchy of human beings or pea from kingdom to species and state the rules of binomial nomenclature.
4	Compare DNA and RNA in any three points. Also state how DNA information is converted into protein.	OR	Draw the general structure of an amino acid, label its four groups, and state one function of proteins.
5	Compare plant and animal cells and state three structural advantages of a plant cell.	OR	Draw and label any one: nucleus, mitochondrion, Golgi apparatus, or endoplasmic reticulum; state the functions of its labelled parts.
6	Differentiate mitosis and meiosis in three points and explain their biological significance.	OR	Identify/describe any stage of mitosis or meiosis from a diagram and state the chromosome-number change involved.
7	Name the epithelial, connective, muscular and nervous tissues in the stomach and give one function of each.	OR	Describe epidermal, mesophyll and vascular tissues of a leaf and relate structure to function.
8	Define enzymes. Compare competitive and non-competitive inhibition with suitable examples or diagrams.	OR	Describe cofactors, coenzymes, activators and prosthetic groups, with examples.
9	Explain why ATP is called the energy currency. Outline glycolysis, Krebs cycle and electron transport chain.	OR	Compare aerobic and anaerobic respiration; distinguish alcoholic and lactic acid fermentation.
10	Explain TACT forces in ascent of sap and the pressure-flow mechanism of food translocation.	OR	Explain effects of temperature, wind and humidity on transpiration and give its advantages/disadvantages.
11	Differentiate macronutrients and micronutrients with examples; explain roles of nitrogen and magnesium.	OR	Explain active and passive absorption of water and mineral salts through roots/root hairs.
12	Describe any three excretory adaptations and any three osmotic adjustments in xerophytes or hydrophytes.	OR	Define homeostasis and explain its importance in plants and humans.
13	Compare natural vegetative propagation and artificial propagation; explain stem cutting and grafting.	OR	Explain binary fission, budding, spore formation and parthenogenesis with examples.
14	Explain development of male gametophyte and the role of the pollen tube.	OR	Explain development of female gametophyte and the process/significance of double fertilization.
15	Explain cloning by plant tissue culture, including explant, callus and plantlet stages.	OR	Describe the structure of a seed and state the fates of ovule, ovary, zygote and primary endosperm nucleus after fertilization.

Core/Reserve guide: Rows 1-10 are the final core prediction. Rows 11-15 cover strong summative alternatives that appeared less consistently but remain examinable.

SECTION C - LONG-QUESTION CORE GUESS

Current pattern: attempt all three questions; choose one side of each OR pair. Each question carries 6 marks.

Q.3	Explain all steps of the scientific method used to solve a biological problem. Show the sequence clearly.
OR	Describe the stages of karyokinesis in mitosis with labelled diagrams, followed by cytokinesis in plant and animal cells.
Q.4	Describe aerobic respiration: glycolysis, formation of acetyl-CoA, Krebs cycle and electron transport chain. Include the overall chemical equation.
OR	Explain light-dependent reactions and the Calvin cycle of photosynthesis with labelled diagrams.
Q.5	Explain development of the female gametophyte and double fertilization in a flowering plant. Draw a labelled embryo-sac diagram.
OR	Draw and describe the internal structure of a leaf, relating epidermal, mesophyll and vascular tissues to their functions.

HIGH-VALUE RESERVE LONG QUESTIONS

Area	Prepare this complete question
Enzymes	Define enzymes; explain their characteristics, mechanism of action, factors affecting activity, and competitive/non-competitive inhibition.
Cell organelles	Draw and explain the structure and functions of nucleus, mitochondrion, Golgi apparatus and endoplasmic reticulum.
Meiosis	Explain Meiosis I with labelled diagrams and show how chromosome number is reduced to half. State its significance in variation.
Plant transport	Explain absorption by roots, ascent of sap through TACT forces, and food translocation by pressure-flow mechanism.
Plant reproduction	Compare natural and artificial vegetative propagation; explain stem, sucker, leaf, bulb, rhizome, tuber, cutting, grafting and tissue culture with diagrams/examples.
Evolution and classification	Explain natural selection, sources of variation and evidence from fossils, homologous structures, vestigial structures and selective breeding.

MCQ ANSWER KEY AND RAPID REVISION

1. B	2. C	3. C	4. B	5. B	6. B	7. C	8. B	9. C	10. C	11. B	12. A
13. A	14. C	15. B	16. B	17. B	18. D	19. B	20. C	21. B	22. B	23. B	24. B
25. B	26. B	27. B	28. B	29. D	30. B	31. B	32. C	33. A	34. B	35. D	36. B

DIAGRAMS THAT MUST BE PRACTISED

Cell biology	Plant cell, animal cell, nucleus, mitochondrion, Golgi apparatus, endoplasmic reticulum, lysosome, vacuole, centriole.
Cell division	Cell cycle; all stages of mitosis; Meiosis I; plant-cell and animal-cell cytokinesis.
Biomolecules	General amino acid, nucleotide, DNA double helix.
Tissues and plant physiology	Cross-section of leaf, root/root hair, stomata, ascent of sap/T-A-C-T, pressure-flow mechanism.
Bioenergetics	Overview of photosynthesis, light reactions, Calvin cycle, aerobic respiration.
Plant reproduction	Bulb, rhizome, tuber, cutting, grafting, tissue culture, pollen grain/male gametophyte, embryo sac/female gametophyte, double fertilization, seed.

LAST-DAY PRIORITY ORDER

1 - Non-negotiable	Scientific method; mitosis/meiosis; organelles; enzymes; respiration/photosynthesis; leaf tissues; sexual reproduction and double fertilization.
2 - Very high value	Evolution evidence; classification/taxonomy; DNA/RNA/protein; plant transport; transpiration; vegetative propagation and tissue culture.
3 - Coverage insurance	Biology careers/interdisciplinary links; stomach tissues; homeostasis; mineral nutrition; excretion and osmotic adjustment.

Full-mark answer habit: answer exactly what the command word asks; use headings; give the required number of points; add labelled diagrams where relevant; compare in a table; and underline key biological terms.